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A Review of MIS Research and Disciplinary Development

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ABSTRACT: The current study focuses on the development of MIS as a field of research and inquiry and examines its intellectual evolution. These issues are addressed through a systematic examination and analysis of 908 MIS articles published between 1968 and 1988 in eight core journals. The articles were analyzed for themes, topics, and research approach. The findings identify popular research topics, the dominant research perspective, and the relationship between MIS research and practice. The popular research topics consist of: IS management, information systems types and characteristics, and development and operation of systems. The dominant research perspective employed in almost all of the empirical articles included in the study can be characterized as a traditional approach reflecting a positivist orientation.

KEY WORDS AND PHRASES: MIS as a discipline, MIS journals, MIS research.

1. Introduction

THE PRIMARY OBJECTIVE OF THE CURRENT STUDY was to develop an overview of the intellectual structure of MIS through direct and systematic analysis of a sample of mainstream MIS articles published in the 1968–88 period. The study aimed at investigating what constitutes the field of MIS by examining topics, themes, and research strategies in the literature. This establishes a benchmark for tracking the status of MIS development and evolution, while focusing attention on areas requiring more research.

The importance of the current study for the still evolving field of MIS lies mainly

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in that it provides an opportunity to pause and reflect upon both what has been accomplished by past work and what needs to be accomplished in the future. As such, it may promote informed discussion and debate and thereby help researchers direct their efforts in the most productive manner. Even emerging fields such as MIS have a sense of history; those that cut themselves off from curiosity and reflection are likely to atrophy [17].

As fields of research or professional practice evolve, they become objects of interest and study themselves [9]. For example, Harvard Business School's 1984 Research Colloquium on Information Systems was devoted to highlighting the research needs of MIS. Furthermore, several studies published during the last decade have reviewed and examined different aspects of the MIS field. Ives, Hamilton, and Davis [13] developed a comprehensive taxonomy of potential MIS research areas and used it to classify over 300 MIS doctoral dissertations. Ives and Olson [14] examined research associated with user involvement in the MIS development process. Jarvenpaa, Dickson, and DeSanctis [15] addressed methodological issues in experimental research. Culnan [6] conducted a co-citation analysis of the MIS literature from 1972 to 1982 in order to identify intellectual subfields in MIS and the reference disciplines within which these subfields are founded. Culnan and Swanson [7] examined published articles from 1980 to 1984 to evaluate the emergence of MIS as an independent scholarly field of study, differentiated from reference disciplines such as computer science, management science, and organizational behavior. Elam, Huber, and Hurt [8] analyzed DSS literature between 1975 and 1985 in order to identify trends in research methodology, topics and application areas. Farhoomand [9] studied the evolution of MIS from a philosophical scientific perspective through examination of published articles in the 1977–85 period. Alavi and Joachimsthaler [1] conducted a meta-analytic review of DSS and individual differences literature published between 1978 and 1988 in order to synthesize and integrate the empirical findings in this area.

The present work builds upon and extends the breadth and depth of the previous studies. For example, in the present study the scope of MIS literature coverage is extended to cover a twenty-year period from 1968 to 1988. Furthermore, instead of being confined to a specific topic or theme (e.g., DSS, GDSS, or implementation) or a specific type of research (e.g., field studies or laboratory experiments), the current study adopted a comprehensive perspective and incorporated research pertaining to any of the environmental, system, or process aspects of MIS.

2. Methodology

EIGHT JOURNALS WERE SELECTED TO BE INCLUDED in the study: *Communications of the ACM, Data Base, Decision Sciences, Harvard Business Review, Journal of Management Information Systems, Management Science, MIS Quarterly,* and *Sloan Management Review.*

The MIS articles published in these journals between 1968 and 1988 were collected and analyzed. For an article to be included in the study, it had to address an aspect of research, management, design, implementation, operation, use, or impact of MIS or

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MIS components. The MIS field draws on several reference disciplines such as computer science, management science, cognitive science, and organizational behavior. Some articles focusing on issues of interest to these disciplines could be considered part of the MIS literature at some level of abstraction. However, such articles were not included in the study unless they were explicitly related to at least one of the above MIS topics. It was felt that analysis of a number of leading or core journals would provide a good understanding of the priorities and issues of interest to the MIS community. The eight journals selected for this study were included in the list of journal preferences of MIS academicians and practitioners for publishing their work [12]. Articles in these journals represent a good mix of MIS scholarly work (e.g., those published in the theory and research section of the *MIS Quarterly* or in the *Journal of Management Information Systems*) and practitioner-oriented work (e.g., those appearing in *Harvard Business Review* and *Sloan Management Review*). Furthermore, most of these journals were included in the previous MIS literature reviews [6, 8, 27]. As such, it was felt that they represent the mainstream of MIS work. All the journals were published in the United States and most had been in publication for at least ten years during the period 1968–88.

3. Analysis

THE JOURNAL ARTICLES INCLUDED IN THE STUDY were classified and coded in three ways:

1. By subject or topic, according to the keyword classification scheme of Barki, Rivard, and Talbot [2].
2. By type, according to a framework developed specifically for the study.
3. By orientation toward research or practice in the field.

After examining some of the existing keyword classification schemes (e.g., *MIS Quarterly* and *Computing Review*), the Barki–Rivard–Talbot classification scheme was adopted and used to code the topic or subject areas addressed by the articles. It was determined that this scheme presents the most comprehensive and current approach to classification of the MIS territory. This classification contains over 1,100 key words and consists of nine top-level categories, each of which is divided into several subcategories. The nine top-level categories are: A—Reference Disciplines; B—External Environment; C—Technological Environment; D—Organizational Environment; E—IS Management; F—IS Development and Operations; G—IS Usage; H—Information Systems; and I—IS Education and Research.

The framework developed for classifying the articles by type is displayed in figure 1. This framework is consistent with the existing MIS methodological taxonomies [24, 26] and extends them. At the highest level, the framework distinguishes between empirical and nonempirical articles. The empirical articles capture the essence of research by relying on observation [26]. The empirical articles are further divided into those that describe an object (e.g., an information system) that endures through time, and those that describe an event or process (e.g., an information system development

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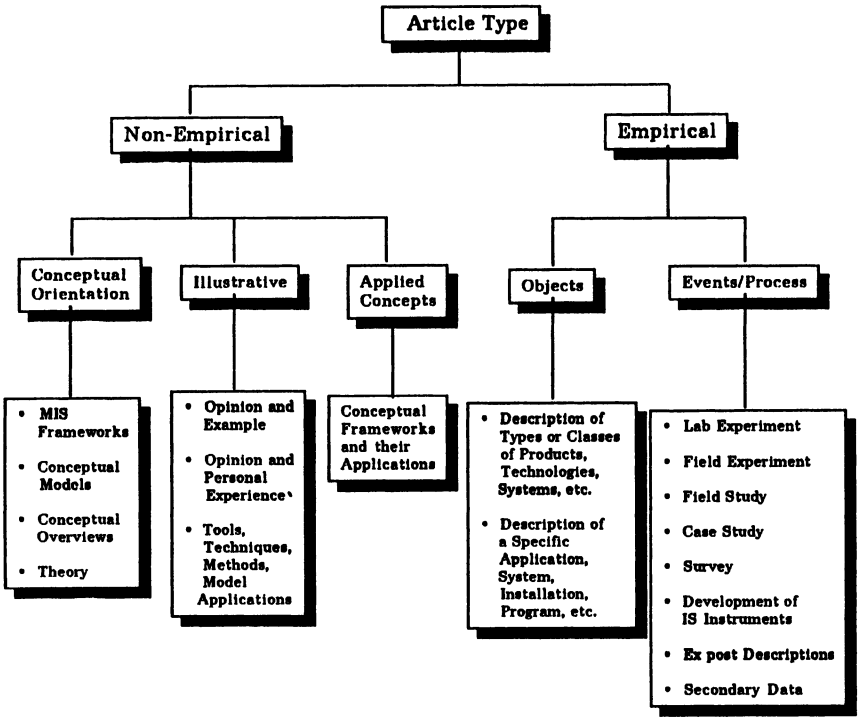

Figure 1 A Framework for Classification of MIS Research Strategies

project) that has a limited life span or is essentially a temporary phenomenon. Empirical articles focusing on objects may describe a system, product, installation, or company MIS function; or they may contain descriptions of types or classes of products, technologies, systems, or projects. Empirical articles focusing on events and processes may take a number of methodological research approaches (e.g., lab experimentation, field experimentation, case study) or they may involve development of MIS instruments (e.g., user satisfaction questionnaire) or design methodologies.

Nonempirical articles are those that are primarily based on ideas, frameworks, and speculation rather than on systematic observation. Nonempirical articles may contain some empirical observations or data, but these will be in a secondary or supporting role only. That is, the emphasis is on ideas rather than on data and observations. Nonempirical articles may be divided into three categories: conceptual, illustrative, and applied concepts. Conceptual articles describe frameworks, models, or theories and offer explanations and reasons. Illustrative articles, on the other hand, are intended to guide practice, often containing recommendations for action or steps to be followed in given circumstances; the stress is on "what" or "how" rather than "why." Applied concepts articles are those that have an approximately equal emphasis on conceptual and illustrative elements. A more detailed description of the classification scheme (displayed in figure 1) and two examples of each article type are provided in the appendix.

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In order to examine the relationship between MIS research and practice, a distinction between research and practitioner-oriented articles was made. Research articles were defined as primarily focused on MIS theory/research and aimed at an academic readership. Practitioner articles were defined as primarily concerned with MIS practice/application. The research and practitioner articles were identified according to the editorial policy of the journals under consideration. Two of the journals, *Harvard Business Review* and *Sloan Management Review* were considered to be predominantly of practitioner orientation. Articles from these two journals were classified as practitioner articles. Similarly, articles in the "Applications" section of *MIS Quarterly* and the "Computing Practices" section of *Communications of the ACM* were classified as practitioner articles. The set of practitioner articles were compared with a set of research articles consisting of articles from *Management Science*, *Decision Sciences*, the "Theory and Research" section of *MIS Quarterly*, and the "Research Contributions" section of *Communications of the ACM*.

4. Procedure

TWO RESEARCHERS WERE INVOLVED IN CODING the articles and worked independently according to the following procedures. The entire collection of the *Journal of Management Information Systems* and *MIS Quarterly* for the time period covered by the study was included. For the six remaining journals (*Communications of the ACM*, *Data Base*, *Decision Sciences*, *Harvard Business Review*, *Management Science*, and *Sloan Management Review*), first, the tables of contents were examined by the coders in order to identify the MIS-related titles. The search was based on the presence of certain key words, including: management, information, computer, data, and system. The tables of contents were then reviewed in a search for recognized MIS authors to identify MIS articles that might have been missed by examination of the article titles alone. The lists of articles identified by each individual coder were pooled to establish a master list. The text of each article in the list was then reviewed in order to exclude articles that might have contained MIS-related key words in the title, but essentially have a non-MIS focus.

Once it was decided to include an article in the study, a coding sheet was used to record data extracted from the articles in order to provide uniformity, consistency, and completeness of data collection. To ensure thoroughness, a short summary was written[1] and the article was coded according to the two classification schemes: the classification scheme for article type shown in figure 1, and the Barki–Rivard–Talbot scheme. Each article was assigned to only one of the classifications shown in figure 1; however, depending on the number of topics covered in the article, more than one Barki code could be assigned to an article. In all, 1,685 Barki codes were assigned to the 908 articles, an average of 1.86 codes per article. The coding sheets were then used to input the data into a dBase III Plus database to facilitate data manipulation and analysis. On the average, each article took about twenty minutes to code.[2]

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5. Results

A TOTAL OF 908 ARTICLES WERE IDENTIFIED AND CODED using the above procedures. The number of articles by journal and by three-year interval is shown in Table 1. As far as non-MIS journals are concerned, 160 articles were published in the practitioner oriented journals (*Harvard Business Review* and *Sloan Management Review*) and 111 articles were published in scholarly journals. As expected, the journals dedicated to MIS or computing and data processing (i.e., *Communications of the ACM, Data Base, Journal of Management Information Systems, MIS Quarterly,*) provided close to 70 percent of the publications.

5.1. Publication Topics

Articles classified in different topic areas of the Barki–Rivard–Talbot classification scheme (one-letter categories A–I) by three-year interval are displayed in Table 2. In the past two decades, the most extensively researched topics in our sample were IS management followed by information systems, and IS development and operations. The specific research issues of interest in each of these three topic areas are shown in Table 3.

In the area of IS management (category E), the three most popular topics were IS evaluation (user satisfaction and cost-benefit analysis), IS planning (IS strategic planning issues), and IS management issues (stage theory). In the information systems usage area (category H), the highest number of articles were written on the topic of information systems types (particularly DSS and expert systems). The next two most popular topics in this area were IS applications and IS characteristics. The highest numbers of articles about IS applications were related to simulation and modeling, government, and banking applications. With regard to IS characteristics area, the main topic of interest has been system interface characteristics, particularly graphics interfaces.

The issues of interest pertinent to category F (IS development and operations) consisted of IS life cycle activities (information requirements analysis and IS design), IS development strategies (prototyping and participative design), and IS implementation (user–analyst differences).

The data displayed in Table 2 indicate that the smallest number of articles in the period 1968–88 were in the topic areas of external environment (category B), technological environment (category C), and organizational environment (category D). The focus of past research in the area of the external environment has been primarily on the economic, social, and legal environments. The articles dealing with the technological environment have primarily addressed database management systems and different aspects of computer hardware systems. The articles published in category D (organizational environment) were primarily focused on organizational structure. The number of published articles addressing the organizational environment issues more than doubled in the early 1980s as compared to the late 1970s.

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Table 1 Frequency of MIS Articles in the Journals by Three-Year Intervals

Journals	1968–70	1971–73	1974–76	1977–79	1980–82	1983–85	1986–88	Total
Communications of the ACM	4	12	7	21	19	42	31	136
Data Base	2	17	6	16	23	32	36	132
Decision Science	1	1	8	5	7	8	12	42
Harvard Business Review	19	11	11	6	19	19	10	95
*Journal of Management Information Systems**	0	0	0	0	0	41	56	97
*MIS Quarterly***	0	0	0	50	64	64	94	272
Management Science	9	6	8	11	12	13	10	69
Sloan Management Review	3	12	9	6	6	14	15	65

* *Journal of Management Information Systems* started publication in 1984. ** *MIS Quarterly* started publication in 1977

Table 2 Articles in Different Topical Areas by Three-Year Intervals

Barki-Rivard-Talbot Codes	Topic Areas	1968–70	1971–73	1974–76	1977–79	1980–82	1983–85	1986–88	Total
E	IS Management	20	21	22	63	74	133	126	459
H	Information Systems	26	29	18	39	57	65	119	353
F	IS Development & Operations	3	6	11	57	44	72	73	266
A	Reference Disciplines	7	19	19	20	32	35	69	201
G	IS Usage	6	1	3	11	12	26	53	112
I	IS Education & Research	2	11	4	9	20	18	23	87
D	Organizational Environment	3	4	2	9	20	24	29	91
C	Technological Environment	1	4	3	6	5	26	28	73
B	External Environment	3	4	7	9	3	12	5	43
Total									1685

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Table 3 Most Popular Research Issues in the Top Three Topic Areas of MIS Research

Barki-Rivard-Talbot Classification Codes	Description	Number of Published Articles
E	IS Management	
EJ	IS evaluation	133
EG	IS planning	74
EM	IS management issues	67
H	Information Systems	
HA	Types of information systems	191
HB	IS application areas	109
HD	IS characteristics	41
F	IS Development and Operations	
FB	IS life cycle activities	98
FA	IS development strategies	65
FD	IS implementation	50

5.2. Article Types

The number of articles by article type and three-year time interval is displayed in figure 2. For each of the three-year intervals between 1968 and 1985, the number of nonempirical articles was greater than the number of empirical articles. Overall, the empirical articles constituted 48.8 percent of the total published between 1968 and 1988. The frequencies of the types of nonempirical articles (conceptual, illustrative, and applied concepts) are displayed in Table 4. With the exception of a peak in the period 1971–73, the number of conceptual articles (MIS frameworks, conceptual models, and theories) has been consistently lower than the number of illustrative articles. A total of twenty-two applied concepts articles were identified. In the empirical articles, shown in Table 5, the most popular research methodologies between 1968 and 1988 were: field studies (used in 146 articles or 33.4 percent of the sample) followed by laboratory experiments (used in 66 or 15.1 percent of the sample). Case studies[3] accounted for 9.1 percent of the empirical articles (a total of 40 articles) and only 4.1 percent of the empirical articles (18 articles) used field experimentation in the period 1968–88. Descriptions of types of products, technologies, etc., and of specific systems, products, installations, etc., accounted for 98 (22.4 percent) of the empirical articles written during this time period.

The number of articles by type and by research topic (Barki–Rivard–Talbot categories A-I) are displayed in figure 3. Interestingly, both empirical and nonempirical articles have primarily focused on the same topics: IS management (category E), information systems (category H), development and operations (category F), and reference disciplines (category A). The smallest numbers of empirical articles appeared in category B (external environment) and category I (IS education and research), with 15 and 17 articles, respectively. The smallest number of nonempirical

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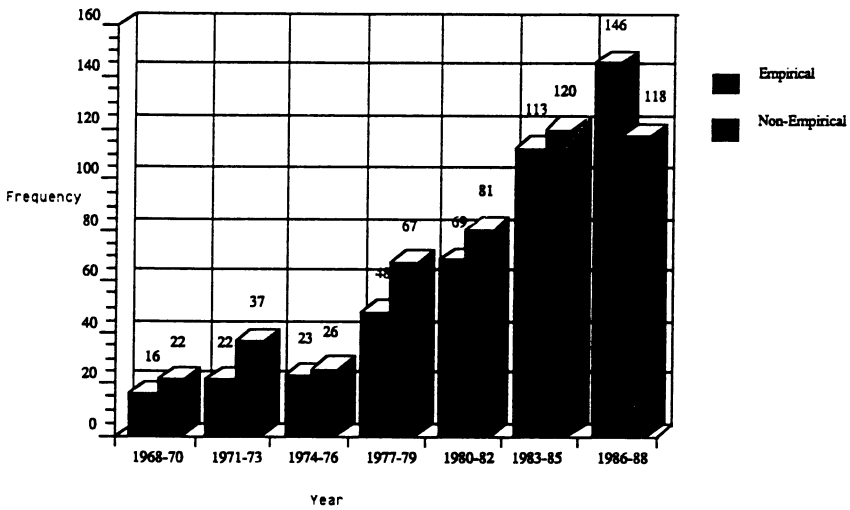

Figure 2 Number of Articles by Type and Year

Table 4 Number of Nonempirical Articles by Type and Three-Year Time Period

Nonempirical articles	1968–70	1971–73	1974–76	1977–79	1980–82	1983–85	1986–88	Total
Conceptual articles	1	24	9	23	29	30	44	160
Illustrative articles	21	13	17	42	47	82	67	289
Applied concepts	0	0	0	2	5	8	7	22
Total								471

articles (29 articles) was published in the category of technological environment (computer systems and software). Relative to empirical articles, a larger number of nonempirical articles (160 versus 199 articles) focused on IS management (category E). On the other hand, more empirical articles addressed the topic of information systems (category H) primarily due to the relatively large number of empirical articles in the area of user-system interface.

5.3. Relationship between MIS Research and Practice

From the total of 679 articles included in this section of the analysis, 376 articles (55 percent) were practitioner-oriented and 303 (45 percent) were research articles. To identify the dominant topics in the research and practitioner articles, we classified the articles according to the Barki–Rivard–Talbot subject categories. The results indicated

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Table 5 Empirical Articles by Type (1968–88)

Research Strategy	Number of Articles
Field Study	146
Lab Experiment	66
Case Studies	40
Survey	32
Field Experiment	18
Other Empirical Type	
Description	98
Ex-post Description	18
Development of tool	12
Secondary Data	7
Total	437

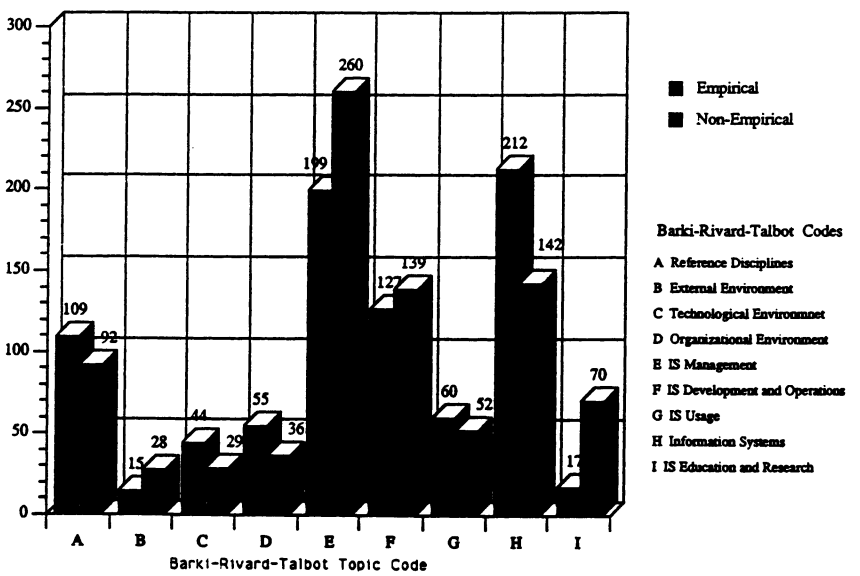

Figure 3 Articles by Topic and Code

that research articles primarily focused on four subject categories: reference disciplines (category A), information systems (category H), IS management (category E), and IS development and operations (category F). The popular subject categories in the practitioner articles (E, H, and F) coincided with three of the dominant subject categories in the research articles. The most popular subject category in the practitioner articles was IS management (category E), which appeared almost twice as many times in the practitioner articles as in the research articles.

We next studied the major themes in the dominant subject categories—E, H, and F. The most frequently occurring theme in category E—IS management—for both research and practitioner articles was IS evaluation. The next most popular themes in

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category E were user satisfaction (dominant in the research articles) and IS planning (dominant in the practitioner articles). The dominant theme in category H—information systems—in the research articles was DSS, while the practitioner articles focused on different application areas. In category F—IS development and implementation—practitioner articles concentrated on development strategies and life cycle activities; while research articles shared the interest in life cycle activities, they also frequently focused on IS implementation themes.

The "origin" of the MIS publication topics was also investigated. If a topic was first published in a practitioner article, it would be of a practitioner origin. On the other hand, if a topic was first published in a research journal, it would have a research/academic origin. The publication dates of the first published research and practitioner articles for the subject categories A–I are displayed in Table 6. Subject categories A and B (reference disciplines and external environment) were of research/academic origin, while subject category E (IS management) was first published in the same year in both research and practitioner articles. For all of the remaining categories (C, D, and F–I), practitioner articles addressed the subject category first.

6. Discussion and Conclusion

BY COLLECTING AND ANALYZING MIS LITERATURE published in core journals in the period 1968–88 we have been able to paint an overall picture of the MIS field and its evolution as represented in this literature.

An analysis of the topic areas covered by the articles indicated a dearth of published work on the topics of external environment (economic, legal, social, and political environments), technological environment (hardware and software), and organizational environment (organization dynamics, structure, and task characteristics). The small number of technological articles represented in the sample may be attributed to the management orientation of the journals included in the study. On the other hand, the research emphasis on the external and organizational environments should be increased. In our opinion, as information systems become more prevalent in organizations and societies, more work on these topics is needed to move the field forward by providing feedback and to guide practice and further research.

Our research reveals a trend in the evolution of the MIS field. Since the mid-1980s there has been a shift in MIS research efforts. For the first time during the past two decades, the number of empirical articles exceeded the number of nonempirical articles essentially focusing on purely conceptual work. This may indicate that the latter type of article was appropriate in the early years of MIS, but with the maturity of the field, there has been a redirection of the research effort paralleled by changes in the publishing standards and practices of core MIS journals. Empirical research articles involving case studies, laboratory experimentation, and field studies have been on the rise during the past twenty years. In particular, the popularity of the laboratory experimentation suggests that an effort is being made to focus on testing theories and/or constructing empirically based theories. However, the small number of theoretically oriented articles (only fifteen in the entire sample) raises questions about the

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Table 6 Publication Dates of the First Articles in Each of the Subject Categories

	Subject Categories								
	A	B	C	D	E	F	G	H	I
Research Articles	1968	1968	1971	1971	1968	1974	1970	1969	1970
Practitioner Articles	1971	1969	1968	1970	1968	1970	1968	1968	1968

rate of progress in the area. We find the paucity of MIS-oriented theories alarming because of the potential implications for quality, experimental design, and value of MIS research. Baroudi and Orlikowski [3] have reported problems with MIS empirical work. They suggest that the average power (the ability to detect treatment effects) of many MIS empirical works is low. In the absence of MIS theories, this can result in important relationships among variables going undetected. In a survey of MIS publications by Jenkins [16], it was concluded that only one-third of the studies chose the appropriate research methodology. Theories are needed to guide research and choice of research methodologies. The impact and role of theories and theoretical views on the choice of research methodologies are addressed by Kling [18] and Weick [28].

Our analysis indicated that MIS research has been closely associated with practice. Almost 300 articles (approximately one-third of all articles published in the period 1968–88) have been devoted to describing and illustrating how the MIS concepts and models could be applied. In fact, relative to purely conceptual/theoretical articles, approximately twice as many illustrative and applied articles have been published in the literature during the past two decades.

Another indication of the close tie between MIS research and practice is that in all but two subject categories, the practitioner interest in the topic coincided with or preceded the research interest. This is an important aspect of the evolution of the field of MIS. It indicates that, while enhancing the MIS research rigor, we have managed to maintain its relevance by focusing on topics of interest and concern to practitioners. It is important to continue to strive for balance between MIS research rigor and relevance in the future.

Almost all of the research approaches employed in the empirical articles included in the study can be characterized as traditional approaches reflecting a positivist orientation. The positivist philosophy of MIS research assumes an objective physical and social world independent of humans that can be characterized and measured. Thus, the researcher assumes the role of an impartial observer who attempts to uncover objective reality through defining constructs and precise measures. The dominance of positivism in MIS research has direct implications for aspects of the phenomena that are studied and for MIS research methodologies. In their assessment of the positivist research philosophy, Orlikowski and Baroudi [24] state that an exclusive view is only a partial view, and the dominance of positivism in MIS research may be unacceptably restrictive. They and other authors [5, 19, 21, 28] have discussed some of the

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restrictions and limitations of the positivist research perspective. For example, the positivist perspective tends to disregard historical and contextual conditions in studying the phenomena of interest, but the development and use of MIS in organizations are intrinsically embedded in social contexts. For this reason, neglecting contextual influences may lead to an incomplete understanding of the phenomenon [24]. Furthermore, the positivist research perspective assumes deterministic and causal relationships underlying the phenomena of interest. As Markus and Robey [20] assert, there is no reason to presume deterministic and nonreciprocal interactions among information systems, people, and organizational settings.

Discussion of the nonpositivist research philosophies is beyond the scope of this paper. However, a number of MIS researchers have started to apply these approaches in the conduct of MIS research. For example, Orlikowski [23] used an interpretive approach to study how the deployment of information technology in primary business activities affected production workers. Smith [25] used a critical research approach to study the impact of the introduction of information technology (in this case, an electronic point of sales system) in retail organizations on retail service workers.

Galliers and Land [11, p. 900] state that, "Traditionally, the topic [IS] has often been viewed as residing for the most part at least, within the province of technology." They further assert that a wider view of IS concerned with social, organizational, and individual factors has emerged, leading to greater complexity, imprecision, and ambiguity. This in turn requires new paradigms for IS research. Thus, methodological pluralism, allowing multiple paradigms for MIS research, should be encouraged [11, 19, 24, 29]. The interpretive perspective, the critical perspective, and the population ecology approaches to MIS research are examples that could open new avenues for gaining insights.

Complementing the dominant positivist MIS research paradigm with alternative perspectives and philosophical bases is important to further development of the field. Progress in the field can be enhanced by adopting a plurality of research perspectives to gain insights into complex MIS phenomena.

NOTES

1. In cases where the coders felt that the article abstract was a good representation of content, the summary was based on the abstract. On the other hand, for the articles for which the abstract was not highly descriptive of the content, the summary was based on material from the article.

2. To calculate intercoder reliability, a random sample of the articles was recoded twice during the coding process. The reliabilities of 71 percent and 70 percent were calculated; these are consistent with the commonly accepted criteria of 70 percent for this type of research.

3. In categorizing the articles, a distinction was made between a case study as a research methodology and an illustrative article that essentially uses exemplars for demonstrative purposes.

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APPENDIX: Classification Scheme for Research Approach

I. Nonempirical Articles

A. Conceptual Orientation

1. MIS Frameworks: proposes a framework for defining the content and scope of MIS, and provides directions for MIS research.

Sprague, R.H. A framework for the development of decision support systems. *MIS Quarterly*, 4, 4 (December 1980), 1–26.

Ives, B.; Hamilton, S.; and Davis, G.B. A framework for research in computer based management information systems. *Management Science*, 20, 9 (September 1980), 910–934.

2. Conceptual Model of a Process or Structure: presents an integrated, schematic representation of MIS-related behavior, activity, organization, method, etc.

King, W.R. Strategic planning for management information systems. *MIS Quarterly*, 2, 1 (March 1978), 27–37.

Gibson, C.F., and Nolan, R.L. Managing the four stages of EDP growth. *Harvard Business Review*, 52, 1 (January–February 1974), 76–88.

3. Conceptual Overview of Ideas, Theories, Concepts, etc.: contains an overview of many concepts or theories in one or more areas, and does not propound or support any individual theory, idea, or approach.

DeSanctis, G. Computer graphics as decision aids: directions for research. *Decision Sciences*, 15, 4 (October 1984), 463–487.

Attewell, P., and Rule, J. Computing and organizations: what we know and what we don't know. *Communications of the ACM*, 27, 12 (December 1984), 1184–1192.

4. Theory from Reference Disciplines: presents theory or theories drawn from outside the MIS field but applied within an MIS context.

Robey, D., and Taggart W. Human information processing in information and decision support systems. *MIS Quarterly*, 6, 2 (June 1982), 61–73.

Huber, G.P. The nature and design of post-industrial organizations. *Management Science*, 30, 8 (August 1984), 928–951.

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B. Illustrative Articles

1. Opinion (Pure, or Supported by Examples): gives advice and guidance for practice, in the form of rules and recommendations, steps or procedures to be followed, hints and warnings, etc. May be supported by examples and applications.

Alavi, M. Managing the risks associated with end-user computing. *Journal of Management Information Systems*, 2, 3 (Winter 1985), 5–20.

Gerrity, T.P., and Rockart, J.F. End-user computing: are you a leader or a laggard? *Sloan Management Review*, 27, 4 (Summer 1986), 25–34.

2. Opinion (Supported by Personal Experience): as for 1, above, but also describes the author's experience in some relevant context (e.g., planning or system development project).

Matlin, G.L. How to survive a management assessment. *MIS Quarterly*, 1, 1 (March 1977), 11–17.

Leonard-Barton, D. The case for integrative innovation: an expert system at Digital. *Sloan Management Review*, 29, 1 (Fall 1987), 7–19.

3. Description of a Tool, Technique, Method, Model, etc.: usually highly specific and detailed, as well as technically or methodologically precise.

Rockart, J.F. Chief executives define their own data needs. *Harvard Business Review*, 57, 2 (March–April 1979), 81–93.

March, S.T., and Carlis, J.V. On the interdependencies between record structure and access path design. *Journal of Management Information Systems*, 4, 2 (Fall 1987), 45–73.

C. Applied Concepts

Conceptual Frameworks and Applications: contains both conceptual and illustrative elements. May present some concept or framework and then describe an application of it.

Naumann, J.D., and Jenkins, A.M. Prototyping: the new paradigm for systems development. *MIS Quarterly*, 6, 3 (September 1982), 29–44.

McFarlan, F.W. Information technology changes the way you compete. *Harvard Business Review*, 62, 3 (May–June 1984), 98–103.

II. Empirical Articles

A. Objects

1. Description of a Type or Class of Products, Technologies, Systems, Projects, etc.

Johansen, R., and Bullen, C. What to expect from teleconferencing. *Harvard Business Review*, 62, 2 (March–April 1984), 164–174.

Vassiliou, Y. On the interactive use of databases: query languages. *Journal of Management Information Systems*, 1, 3 (Winter 1984), 22–48.

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2. Description of a Specific Application System, Product, Installation, Software Model, Program, Company, IS Function, etc.

Hall, W.A., and McCauley, R.E. Planning and managing a corporate network utility. *MIS Quarterly*, 11, 4 (December 1987), 437–449.

Bonczek, R.H.; Holsapple, C.W.; and Whinston, A.B. Aiding decision makers with a generalized data base management system: an application to inventory management. *Decision Sciences*, 9, 2 (April 1978), 228–245.

B. Events/Process

1. Laboratory Experiment: manipulates independent variable; controls for intervening variables; conducted in controlled settings.

Alavi, M., and Henderson, J.C. An evolutionary strategy for implementing a decision support system. *Management Science*, 27, 11 (November 1981), 1309–1323.

Benbasat, I.; Dexter, A.S.; and Masulis, P.S. An experimental study of the human/computer interface. *Communications of the ACM*, 24, 11 (November 1981), 752–762.

2. Field Experiment: as for laboratory experiment, but in a natural setting of the phenomenon under study.

Franz, C.R.; Robey, D.; and Koeblitz, R.R. User response to an online information system: a field experiment. *MIS Quarterly*, 10, 1 (March 1986), 29–42.

Barber, R.E., and Lucas, H.C. System response time, operator productivity and job satisfaction. *Communications of the ACM*, 26, 11 (November 1983), 972–986.

3. Field Study: no manipulation of independent variables, involves experimental design but no experimental controls, is carried out in the natural settings of the phenomenon of interest.

Couger, J.D., and McIntyre, S.C. Motivation norms of knowledge engineers compared to those of software engineers. *Journal of Management Information Systems*, 4, 3 (Winter 1987), 82–93.

Swanson, E.B. Information channel disposition and use. *Decision Sciences*, 18, 1 (January 1987), 131–145.

4. Case Study:

a. Single Case: examines a single organization, group, or system in detail; involves no variable manipulation, experimental design or controls; is exploratory in nature.

Harrison, W. Using software matrices to allocate testing resources. *Journal of Management Information Systems*, 4, 4 (Spring 1988), 91–105.

Kling R., and Iacono, S. The control of information systems development after implementation. *Communications of the ACM*, 27, 12 (December 1984), 1218–1226.

b. Multiple Case Studies: as for single case studies, but carried out in a small number of organizations or context.

Olson, M.H. Remote office work: changing work patterns in space and time. *Communications of the ACM*, 26, 3 (March 1983), 182–187.

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Boynton, A.D., and Zmud, R.W. An assessment of critical success factors. *Sloan Management Review*, 25, 4 (Summer 1984), 17–27.

5. Survey: involves large numbers of observations; the research uses an experimental design but no controls.

Adams, C. How management users view information systems. *Decision Sciences*, 6, 2 (April 1975), 337–345.

Dickson, G.S.; Leitheiser, R.L.; Wetherbe, J., et al. Key information systems issues for the 1980's. *MIS Quarterly*, 8, 3 (September 1984), 135–159.

6. Description of the Development of Instrument/Measurements or Classification Scheme.

Bailey, J.E., and Pearson, S.W. Development of a tool for measuring and analyzing computer user satisfaction. *Management Science*, 29, 5 (May 1983), 530–545.

Barki, H.; Rivard, S.; and Talbot, J. An information systems keyword classification scheme. *MIS Quarterly*, 12, 2 (June 1988), 299–322.

7. Ex-Post Description of Some Project or Event: interest in reporting the results of the project develops after the project is complete (or is partially complete).

Kozar, K.A., and Mahlum, J.M. A user generated information system: an innovative development approach. *MIS Quarterly*, 11, 2 (June 1987), 163–174.

Hiltz, S.R., and Turoff, M. The evolution of user behavior in a computerized conferencing system. *Communications of the ACM*, 24, 11 (November 1981), 739–751.

8. Secondary Data: research using data from secondary sources, that is, data collected by sources other than the researcher.

Dutton, W.H., and Kraemer, K.L. Management utilization of computers in American local governments. *Communications of the ACM*, 21, 3 (March 1978), 206–218.

Mendelson, H. Economies of scale in computing: Grosch's law revisited. *Communications of the ACM*, 30, 12 (December 1987), 1066–1072.

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